

ESP32 Based Microcontroller.

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This project involved the design and development of a custom microcontroller board based on the Espressif ESP32-C3. The board integrates an Inertial Measurement Unit (IMU) capable of measuring linear acceleration and angular velocity for motion sensing applications. This section describes the system design, development process, and key engineering considerations.

The ESP32-C3 supports multiple boot modes, which are determined by the state of specific GPIO pins during startup. For USB-to-Serial programming, GPIO2 and GPIO8 are pulled high using resistors, while GPIO9 is connected to ground through a push-button switch. Pressing the switch during reset places the device in bootloader mode for firmware upload.

A manual reset function is implemented by connecting the EN (reset) pin to a push-button switch that pulls the line low when pressed. It is also to be noted that external peripherals can be connected through the exposed I2C, SPI and UART pins. The complete schematic of the board is shown in Figure 1.

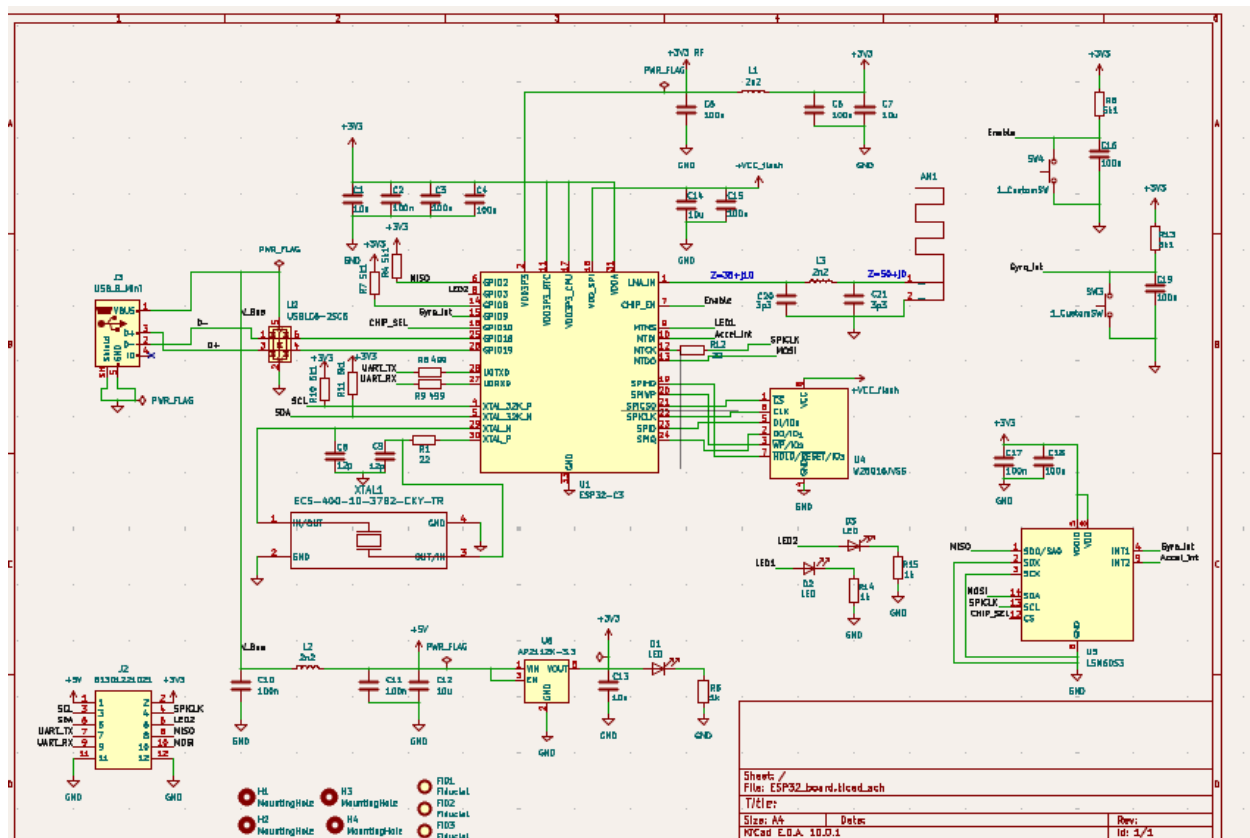


Figure 1: Overall Schematic

Power

The ESP32-C3 requires a regulated 3.3 V supply for operation. Power is provided through a USB Mini connection and converted to 3.3 V using a voltage regulator. To ensure stable operation, multiple decoupling capacitors are incorporated throughout the design to suppress voltage fluctuations and high-frequency noise. In addition, ESD protection is included on the USB interface to protect the board from electrostatic discharge events.

A power indicator LED is connected to the regulator output to provide a visual indication of proper power conversion. To further improve power integrity, a π -filter is placed between the USB input and the regulator to attenuate noise originating from both the USB source and the power conversion circuitry. Along with a 10 μ F bulk decoupling capacitor, a 100 nF decoupling capacitor is placed near each power pin to minimize local noise and maintain supply stability. The schematic of the 3.3 V power supply circuit is shown in Figure 2.

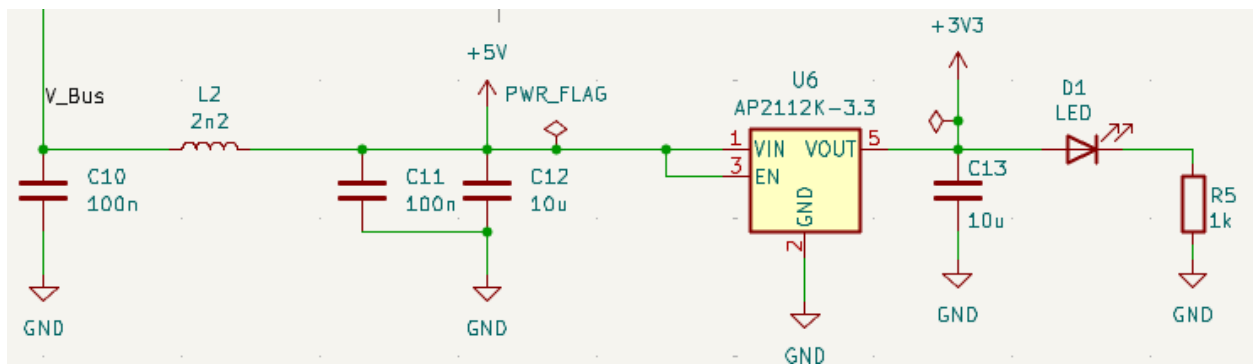


Figure 2: 5V to 3.3V voltage regulator

Timing

A 40 MHz crystal oscillator is used as the system clock source for the ESP32-C3. The crystal has a specified load capacitance of 10 pF, and the PCB stray capacitance is estimated to be 4 pF. Using the method described in the STMicroelectronics application note AN2867, the required external load capacitance is calculated as:

$$C_{\text{load}} = 2(10 \text{ pF} - 4 \text{ pF}) = 12 \text{ pF}$$

Therefore, 12 pF capacitors are connected to each side of the crystal to achieve the desired load capacitance. Additionally, a 22 Ω series resistor is placed at the oscillator output to limit the drive strength and improve signal integrity. The oscillator circuit schematic is shown in Figure 3.

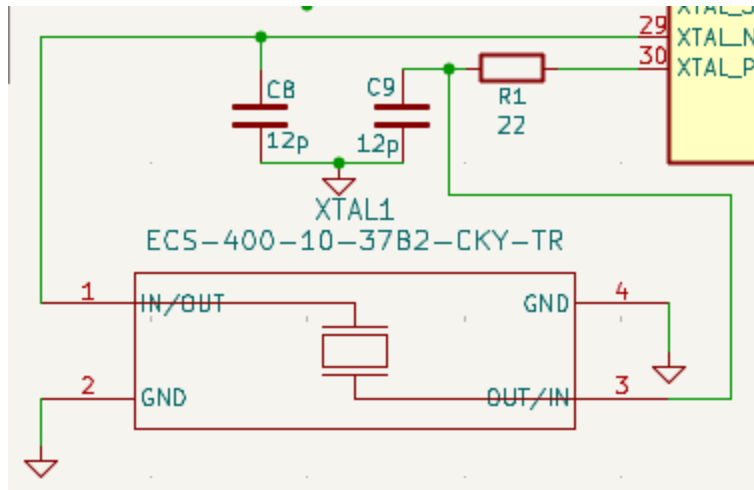


Figure 3: Crystal Oscillator Connection

Flash Memory

The ESP32-C3 requires an external flash memory device for firmware storage and program execution. For this design, the W25Q16JVSS serial flash memory IC was selected due to its compatibility with the ESP32-C3 and sufficient storage capacity.

Communication between the ESP32-C3 and the flash memory is implemented using the SPI interface. The flash memory is connected directly to the dedicated SPI pins specified by Espressif, resulting in a simple and reliable hardware implementation. The schematic showing the connections between the ESP32-C3 and the flash memory is presented in Figure 4.

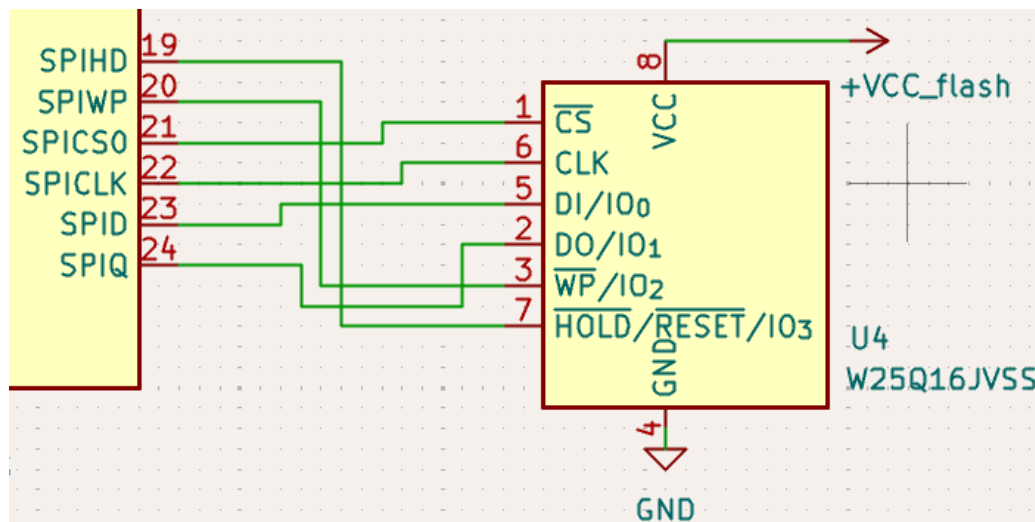


Figure 4: Flash memory

Inertial Measurement Unit (IMU)

The board incorporates an LSM6DS3 IMU from STMicroelectronics, which integrates a three-axis accelerometer and three-axis gyroscope for motion sensing. Although the device supports both I²C and SPI communication, SPI was selected due to its simplicity and compatibility with the available ESP32-C3 pins. The IMU interrupt pins are also connected to the ESP32-C3, allowing sensor events to trigger interrupts when required. Figure 5 shows the connections between the IMU and the ESP32-C3.

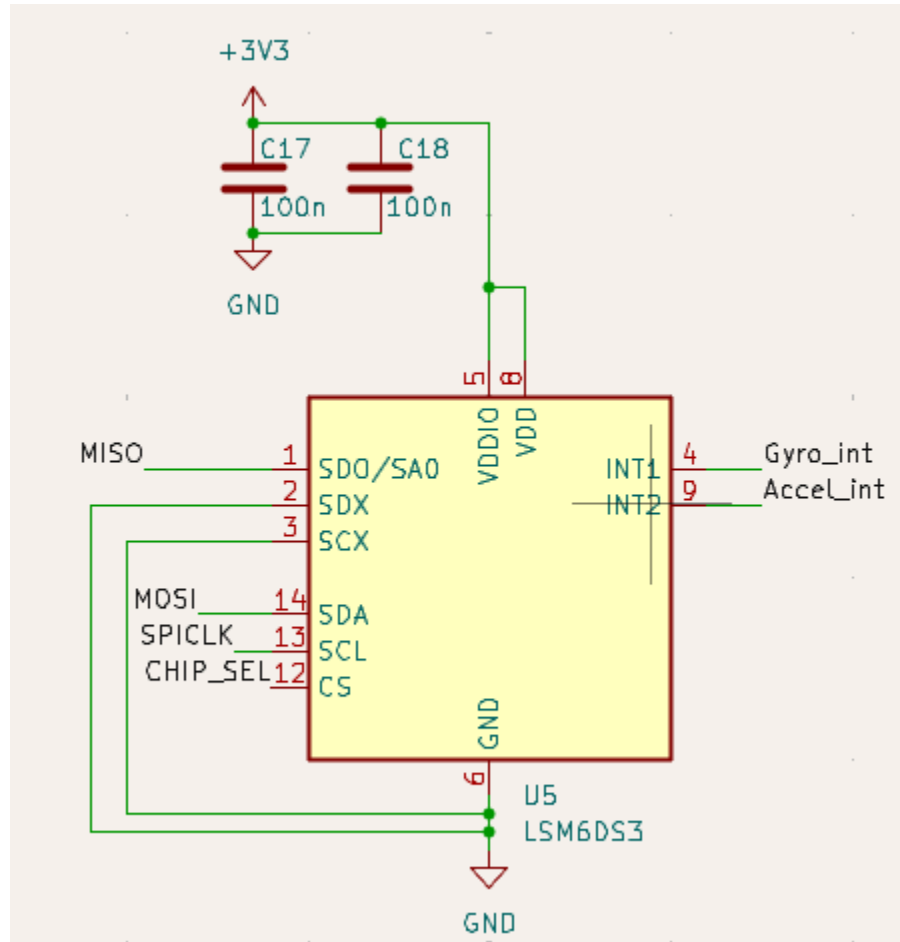


Figure 5: Connection between ESP-32 and IMU

PCB Antenna

The ESP32-C3 includes integrated Wi-Fi and Bluetooth capabilities, which require a 2.4 GHz antenna for wireless communication. To support this functionality, a 2.4 GHz Inverted-F Antenna was implemented based on the design guidelines provided in Texas Instruments Application Note AN043. The antenna is designed for a characteristic impedance of 50 Ω , while the ESP32-C3 RF input has an impedance of approximately $35 + j10 \Omega$. To maximize power transfer and minimize signal reflections, a π -matching network consisting of capacitors and an inductor

was designed using an impedance-matching calculator. Figure 6 shows the connections between the antenna and the ESP32-C3.

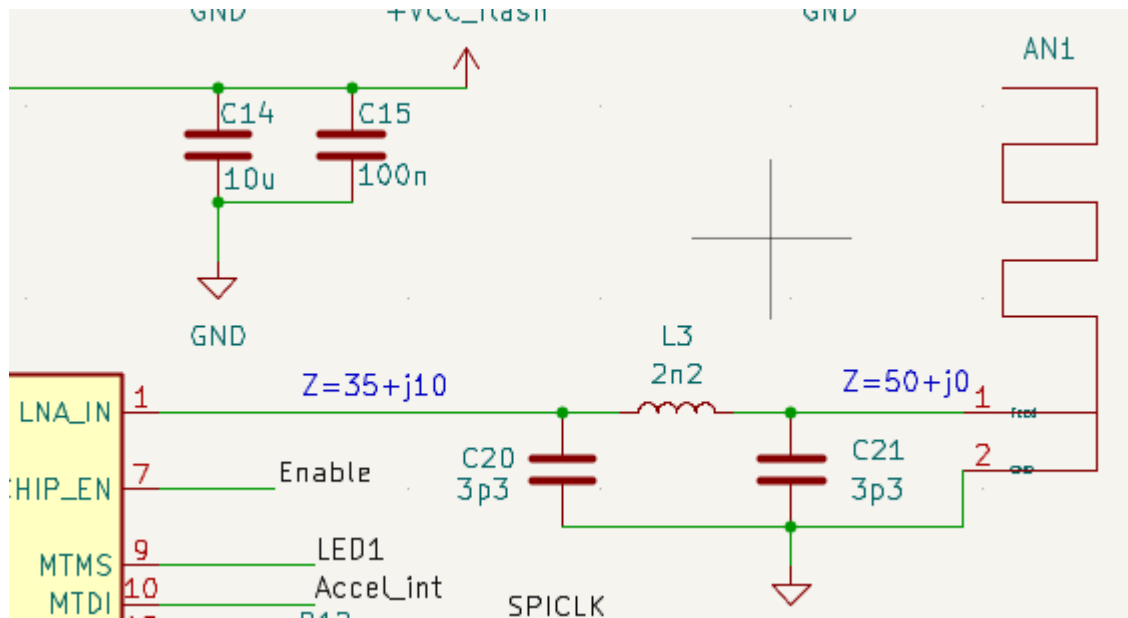


Figure 6: PCB antenna connection

PCB Routing and Manufacturing

Figure 7 shows the PCB routed using KiCad. Figure 8 shows the 3D model of the PCB. This is a 4 layer PCB where the stackup is as follows:

Table 1: PCB stackup

Layer Number	Layer Function
Layer 1	Signals
Layer 2	Ground
Layer 3	Used to route power signals
Layer 4	Signals

During PCB layout, through-hole, blind, and buried vias were initially used to route signals between layers while maintaining a compact design. Decoupling capacitors were placed as close as possible to their corresponding power pins to minimize noise, and the crystal oscillator was positioned away from high-speed signals to reduce timing interference. Following the recommendations of Texas Instruments Application Note AN043, the PCB antenna was placed at the edge of the board with the required clearance from surrounding components.

After completing the layout, Gerber files were generated for manufacturing. However, many low-cost PCB manufacturers, such as JLCPCB, do not support blind and buried vias, and manufacturers that do offer this capability typically charge significantly higher fabrication costs. Therefore, the design will be revised to use only through-hole vias, resulting in a more cost-effective and manufacturable PCB.

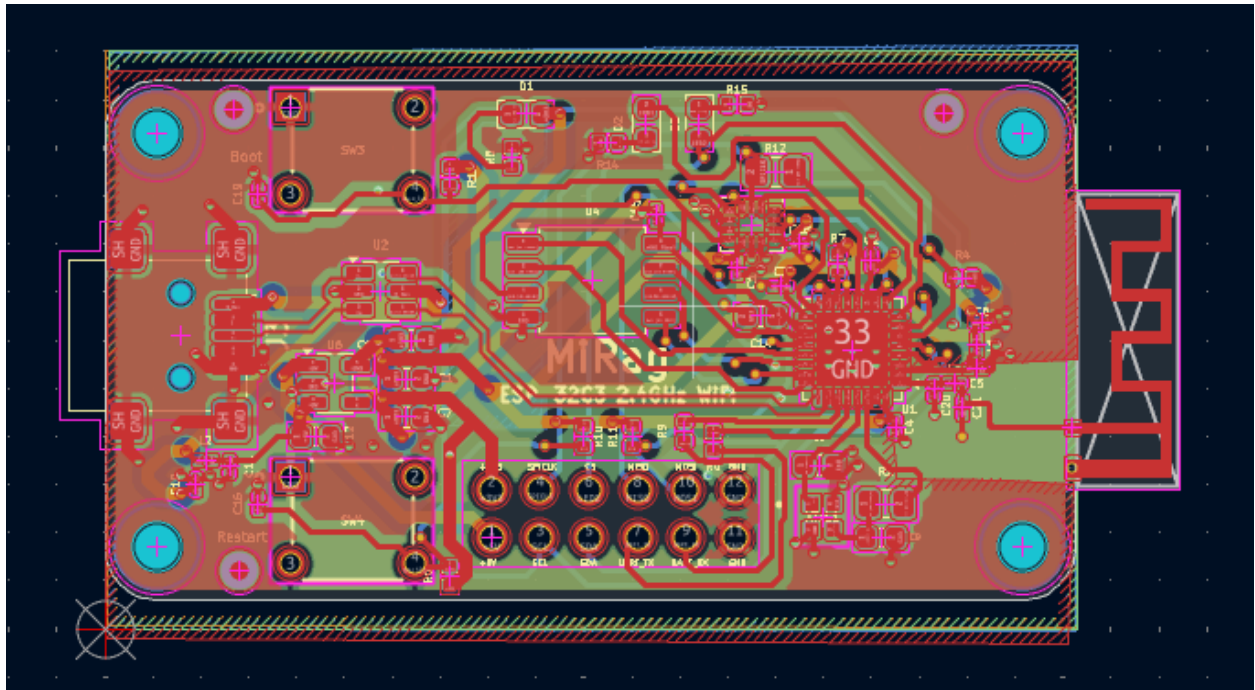


Figure 7: PCB routing

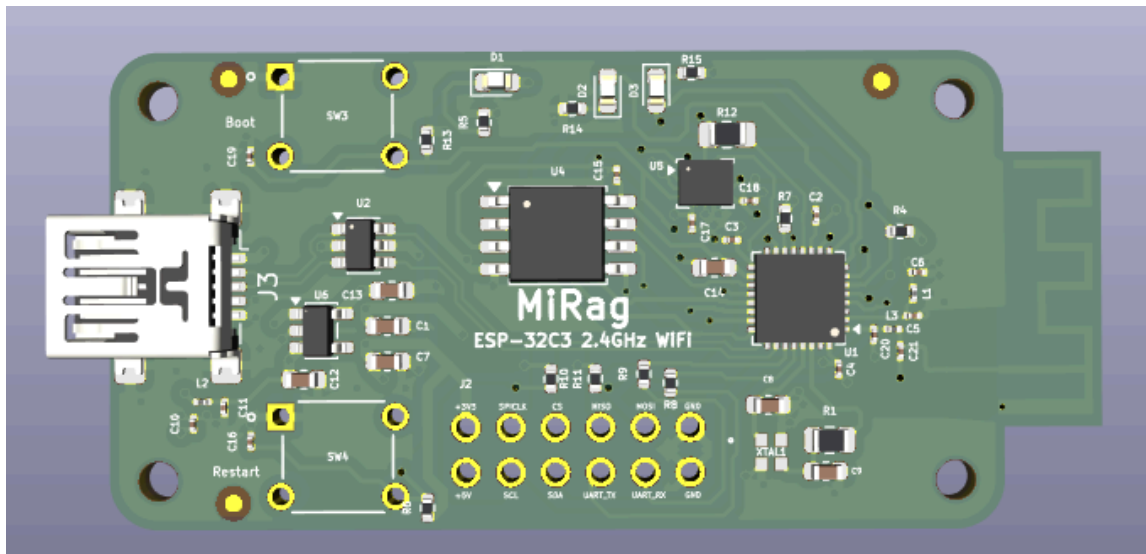


Figure 8a: 3D view of the Microcontroller board

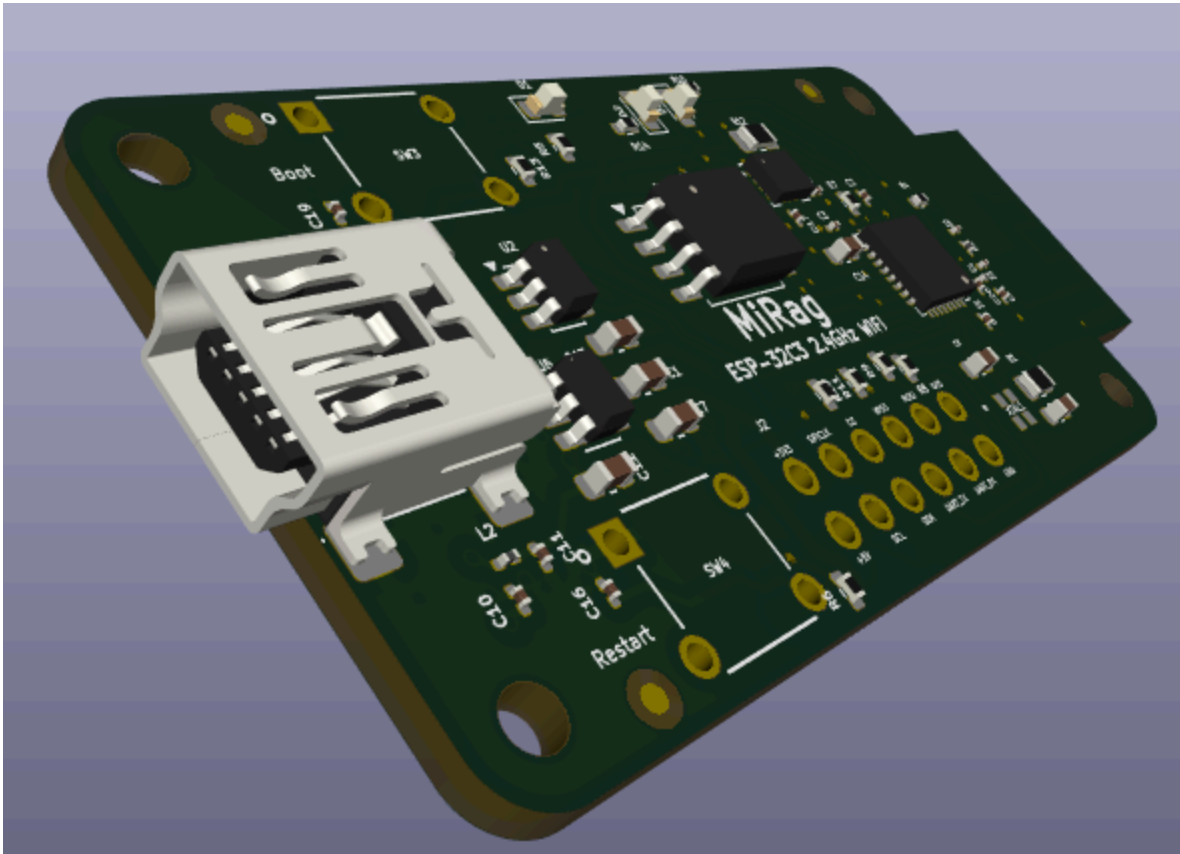


Figure 8b: 3D view from USB Mini side